

CellCast – AI Forecasting a Cell's Future Before the Experiment

CellMind Team

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Abstract:

Single-cell RNA sequencing has become an important technology for studying cellular responses to external perturbations. However, performing perturbation experiments across many cell types, conditions, and species is costly and time-consuming. This study focuses on AI-assisted single-cell perturbation response prediction, aiming to predict perturbed gene expression from control-state cells and reduce the experimental burden of perturbation biology. We propose CellCast, an axial-attention-based deep learning model for single-cell perturbation prediction. CellCast represents full-gene expression profiles as a fixed computational gene grid and uses row-wise and column-wise attention to model gene-token dependencies. Instead of directly generating perturbed profiles, CellCast predicts perturbation-induced residual gene-expression changes, making the prediction process consistent with the biological idea that perturbations modify an existing cellular state. We evaluated CellCast on two tasks: i.i.d. prediction on a COVID-19 perturbation dataset and cross-species prediction on an LPS6 dataset. On the COVID-19 dataset, CellCast achieved the best average mean-expression R^2 of 0.9716 across six cell types and improved response error, DEG recovery, and response direction prediction. In the cross-species experiment, CellCast achieved an average R^2 of 0.9982 across mouse, rat, rabbit, and pig, and recovered more target-species response genes. These results show that CellCast can capture population-level perturbation responses and generate useful computational hypotheses for biological experiments. The study provides an innovative AI-assisted framework for prioritizing perturbation conditions and response genes before costly laboratory validation.

 **Project Page:** <https://grace-lilok.github.io/CellCast>

1 Introduction & Research Objectives

Single-cell RNA sequencing (scRNA-seq) has greatly advanced biomedical research by enabling transcriptome profiling at single-cell resolution [Shalek et al. \(2013\)](#); [Regev et al. \(2017\)](#). It has been widely used to characterize cellular heterogeneity, identify cell types and states, and investigate dynamic transcriptional changes under different biological conditions. Among these applications, understanding how cells respond to perturbations, such as viral infection, cytokine stimulation, drug treatment, and gene perturbation, is particularly important for studying disease mechanisms, therapeutic responses, and cellular state transitions [Dixit et al. \(2016\)](#); [Ji et al. \(2021\)](#).

However, systematic single-cell perturbation experiments remain expensive and difficult to scale across cell types, perturbation conditions, doses, and time points. In addition, scRNA-seq measurements are usually destructive, meaning that the same cell cannot be measured both before and after perturbation. Therefore, perturbation datasets often consist of unpaired control and perturbed

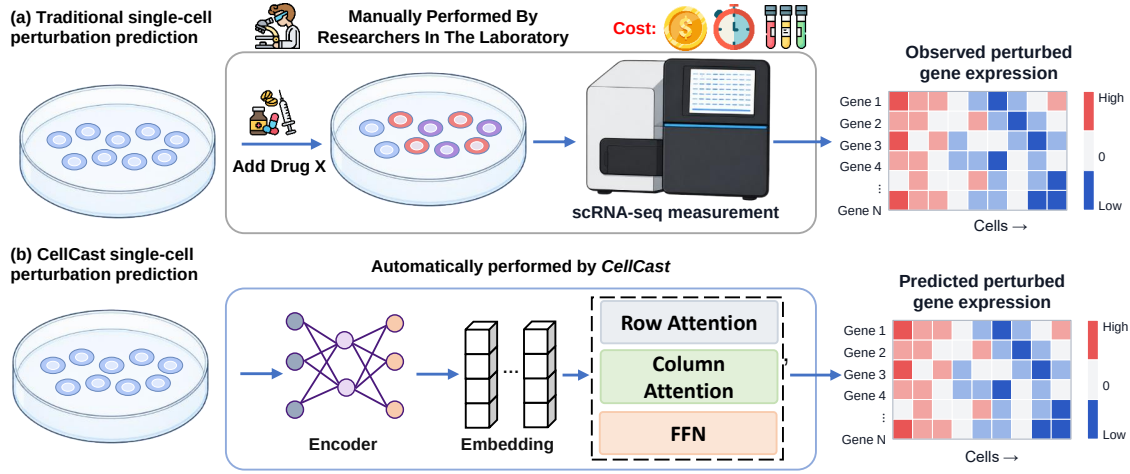


Figure 1 AI-assisted single-cell perturbation prediction. (a) Conventional perturbation profiling relies on laboratory treatment and scRNA-seq measurement to obtain observed perturbed gene expression. (b) CellCast uses AI to predict perturbed gene expression directly from control-state single-cell profiles through gene-token embedding and axial attention.

cell populations. These challenges have motivated the development of computational models for *in silico* single-cell perturbation prediction, whose goal is to predict perturbed gene expression profiles from control-state expression profiles. As illustrated in Figure 1, such models provide an AI-assisted alternative to conventional laboratory-based perturbation profiling by predicting perturbation-induced transcriptional responses directly from control single-cell expression data.

Several computational methods have been proposed for this task. Early statistical models, such as regularized linear models, provide interpretable estimates of perturbation effects but may be limited in modeling nonlinear cellular responses [Dixit et al. \(2016\)](#). Deep generative models have further improved perturbation prediction by learning latent representations of cellular states. For example, scGen predicts perturbation responses through latent vector arithmetic [Lotfollahi et al. \(2019\)](#), trVAE models conditional out-of-distribution generation [Lotfollahi et al. \(2020\)](#), scPreGAN uses adversarial learning for perturbation translation [Wei et al. \(2022\)](#), and CoupleVAE learns nonlinear transformations between control and perturbed latent spaces using coupled variational autoencoders [Wu et al. \(2025\)](#). Other methods, such as CPA, CellIOT, and GEARS, model compositional perturbations, optimal transport-based distribution mapping, or genetic perturbation effects with prior knowledge [Lotfollahi et al. \(2023\)](#); [Bunne et al. \(2023\)](#); [Roohani et al. \(2024\)](#).

Despite these advances, many existing methods mainly focus on global cellular state transition, while gene-wise perturbation responses are often modeled only implicitly in latent space. In many biological analyses, however, recovering the direction and magnitude of gene-level responses is crucial, especially for identifying differentially expressed genes and interpreting perturbation-induced transcriptional programs. Moreover, directly modeling interactions among thousands of genes is computationally challenging, making it difficult to explicitly capture full-gene dependencies in high-dimensional scRNA-seq data.

To address these limitations, we propose **CellCast**, a full-gene axial-attention model for single-cell perturbation response prediction. Instead of directly generating perturbed expression from scratch, CellCast formulates perturbation prediction as a residual gene-expression response prediction problem: $\hat{x}^{pert} = x^{ctrl} + \Delta x$, where (Δx) denotes the perturbation-induced gene expression response predicted from the control cell. CellCast organizes the full gene expression vector into a

fixed computational gene grid and applies row-wise and column-wise axial attention to efficiently model gene-token dependencies. To distinguish genes with identical expression values, CellCast further combines expression projection, learnable gene identity embeddings, and two-dimensional positional embeddings to construct gene-aware token representations.

In summary, the main contributions of this work are three-fold. First, we formulate single-cell perturbation prediction as residual gene-expression response prediction, enabling the model to explicitly focus on perturbation-induced changes. Second, we introduce a full-gene axial-attention architecture for efficient gene-token dependency modeling without relying on predefined gene rankings, DEG filtering, pathway annotations, or gene regulatory networks. Third, we evaluate perturbation prediction from a response-oriented perspective, including global expression accuracy, perturbation delta recovery, differentially expressed gene recovery, and response direction consistency.

2 Methods

2.1 Problem Formulation

Let $x^{ctrl} \in \mathbb{R}^G$ denote the gene expression profile of a single cell under the control condition, where G is the number of genes. The goal of single-cell perturbation prediction is to predict the corresponding expression profile under a perturbed condition, denoted as $\hat{x}^{pert} \in \mathbb{R}^G$.

Given control and perturbed single-cell populations under matched perturbation settings, CellCast learns a mapping function

$$f_\theta : x^{ctrl} \rightarrow \hat{x}^{pert}. \quad (1)$$

Since single-cell perturbation datasets are usually composed of unpaired control and perturbed cell populations, the model does not assume that the same cell is observed before and after perturbation. Instead, it learns to map control-state expression profiles to the corresponding perturbed expression distribution under matched cell types or perturbation conditions.

Rather than directly generating the perturbed expression profile from scratch, we formulate perturbation prediction as a residual gene-expression response prediction problem:

$$\hat{x}^{pert} = x^{ctrl} + \Delta x, \quad (2)$$

where $\Delta x \in \mathbb{R}^G$ represents the perturbation-induced gene expression response predicted by the model. This formulation reflects the biological intuition that perturbations usually modify the original cellular state rather than completely overwrite it.

2.2 Overview of CellCast

CellCast is a full-gene axial-attention model for single-cell perturbation response prediction. As shown in Figure 2, CellCast formulates perturbation prediction as a residual gene-expression response prediction problem. Given a control expression profile, CellCast first organizes the full gene expression vector into a fixed two-dimensional computational gene grid. Each gene is then represented as a gene-aware token by combining expression projection, learnable gene identity embeddings, and two-dimensional positional embeddings. The resulting gene grid is processed by stacked axial attention blocks, which model gene-token dependencies along row and column axes. Finally, an output head predicts the residual perturbation response Δx , and the predicted perturbed expression is obtained by adding Δx back to the original control expression.

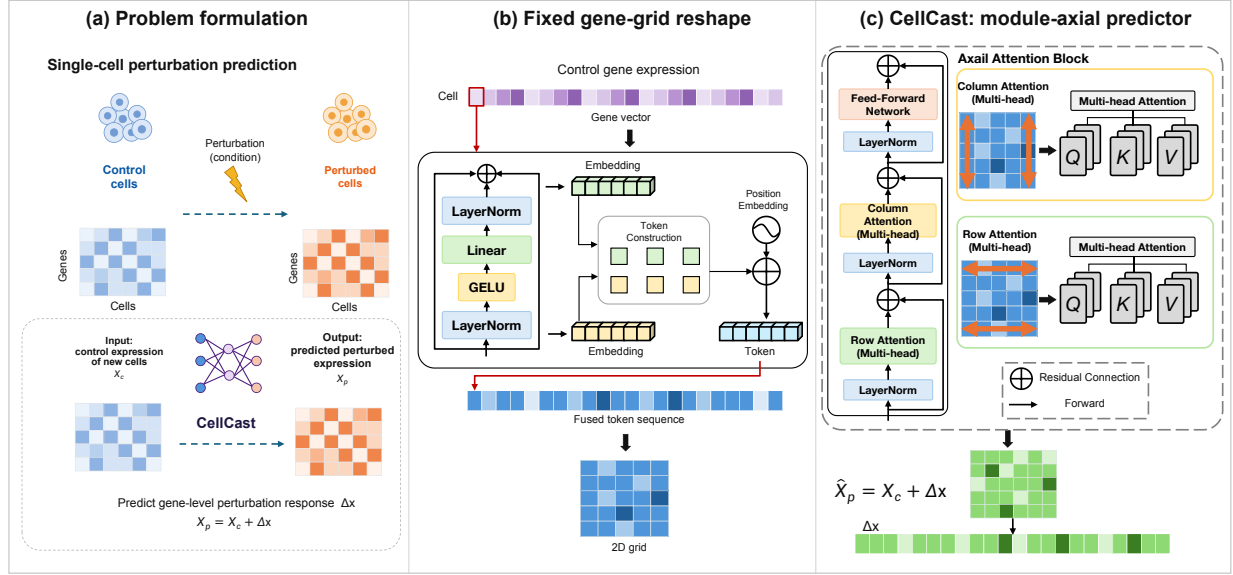


Figure 2 Architecture of CellCast for single-cell perturbation prediction. (a) CellCast predicts perturbed expression from control cells by estimating gene-level perturbation responses. (b) Gene expression vectors are embedded and reshaped into a fixed two-dimensional gene grid. (c) The module-axial predictor uses row-wise and column-wise attention blocks to model gene-token dependencies and predict the residual response Δx .

Specifically, Figure 2(a) illustrates the overall problem formulation of predicting perturbed expression from control cells. Figure 2(b) shows how the one-dimensional gene expression vector is embedded and reshaped into a fixed two-dimensional gene grid. Figure 2(c) presents the module-axial predictor, where row-wise and column-wise attention blocks are used to model gene-token dependencies and predict the perturbation response.

The overall prediction process can be summarized as

$$x^{ctrl} \rightarrow \text{Gene Tokenization} \rightarrow \text{Axial Attention} \rightarrow \Delta x \rightarrow \hat{x}^{pert}. \quad (3)$$

2.3 Gene Grid Construction

For each cell, the input is represented as a full gene expression vector:

$$x^{ctrl} = [x_1, x_2, \dots, x_G] \in \mathbb{R}^G, \quad (4)$$

where G denotes the number of genes. To make axial attention applicable to high-dimensional gene expression data, CellCast reshapes the one-dimensional gene vector into a fixed two-dimensional computational gene grid, as illustrated in Figure 2(b). Let H and W denote the height and width of the grid, satisfying

$$H \times W \geq G. \quad (5)$$

If $H \times W > G$, zero padding is added to the end of the gene vector, and the padded vector is reshaped into a two-dimensional grid $X \in \mathbb{R}^{H \times W}$. The gene grid is used only as a computational structure for axial attention. It does not represent biological spatial organization, gene regulatory networks, biological pathways, or chromosomal positions. The same fixed gene order is used during both training and inference.

2.4 Gene Tokenization

After constructing the gene grid, CellCast converts each gene into a token representation. Each scalar gene expression value x_i is first projected into a hidden representation:

$$p_i = W_p x_i + b_p, \quad (6)$$

where p_i is the projected expression representation. Since different genes may have identical or similar expression values, expression projection alone cannot distinguish gene identities. Therefore, CellCast introduces a learnable gene identity embedding and two-dimensional positional embeddings. The final token representation of the i -th gene is defined as

$$h_i = p_i + e_i + r_{\text{row}(i)} + c_{\text{col}(i)}, \quad (7)$$

where e_i is the learnable gene identity embedding, and $r_{\text{row}(i)}$ and $c_{\text{col}(i)}$ are the row and column positional embeddings, respectively. The resulting tokens form the initial gene-grid representation H^0 .

2.5 Axial Attention Gene Response Module

Directly applying full self-attention to thousands of genes is computationally expensive. To address this problem, CellCast uses axial attention, which decomposes full gene-token interaction modeling into row-wise and column-wise attention operations [Vaswani et al. \(2017\)](#); [Ho et al. \(2019\)](#). As shown in Figure 2(c), row-wise attention first models dependencies among gene tokens within each row of the gene grid:

$$\bar{H}^l = \text{RowAttn}(H^l). \quad (8)$$

Column-wise attention is then applied to integrate information across rows:

$$\tilde{H}^l = \text{ColAttn}(\bar{H}^l). \quad (9)$$

The output is further transformed by a feed-forward network:

$$H^{l+1} = \text{FFN}(\tilde{H}^l). \quad (10)$$

Each axial attention block also includes residual connections and layer normalization to stabilize training. By alternating row-wise and column-wise attention, CellCast allows information to be exchanged across the full gene grid while avoiding the high computational cost of full self-attention over all genes.

2.6 Residual Perturbation Response Prediction

After stacked axial attention blocks, CellCast obtains contextualized gene-token representations. An output head maps these representations to a gene-level perturbation response vector Δx . Instead of directly generating the perturbed expression profile from scratch, CellCast predicts the residual response and computes the final perturbed expression as

$$\hat{x}^{\text{pert}} = x^{\text{ctrl}} + \Delta x. \quad (11)$$

This residual prediction strategy is consistent with the biological intuition that perturbations usually modify the original cellular state rather than completely overwrite it. Therefore, the model is encouraged to focus on perturbation-induced gene expression changes.

2.7 Training Objective

CellCast is trained by comparing the predicted perturbed expression $\hat{x}^{pert}$ with the observed perturbed expression x^{pert} . The primary prediction loss is defined as

$$\mathcal{L}_{pred} = \frac{1}{N} \sum_{i=1}^N \left\| \hat{x}_i^{pert} - x_i^{pert} \right\|_2^2, \quad (12)$$

where N denotes the number of training samples.

To further encourage the model to recover population-level perturbation responses, we introduce a mean response consistency loss:

$$\mathcal{L}_{mean} = 1 - \text{PCC} \left(\mu(\hat{X}^{pert}), \mu(X^{pert}) \right), \quad (13)$$

where $\mu(\cdot)$ denotes the mean expression profile across cells and PCC denotes the Pearson correlation coefficient.

The final training objective is

$$\mathcal{L} = \mathcal{L}_{pred} + \lambda \mathcal{L}_{mean}, \quad (14)$$

where λ controls the contribution of the mean response consistency loss.

2.8 Evaluation Metrics

We evaluate CellCast from both expression-level and response-level perspectives. Expression-level metrics measure the similarity between predicted and observed perturbed expression profiles, including mean squared error, mean absolute error, Pearson correlation, Spearman correlation, and R^2 .

To evaluate whether the model captures perturbation-induced gene expression changes, we compute the real and predicted perturbation responses:

$$\Delta^{real} = \mu(X^{pert}) - \mu(X^{ctrl}), \quad \Delta^{pred} = \mu(\hat{X}^{pert}) - \mu(X^{ctrl}). \quad (15)$$

The agreement between Δ^{real} and Δ^{pred} is assessed using response correlation, response direction accuracy, and differentially expressed gene recovery. These response-oriented metrics help determine whether CellCast not only predicts the final perturbed expression profile, but also captures biologically meaningful perturbation responses.

2.9 Experimental Design

We designed experiments under two settings: independent and identically distributed (i.i.d.) prediction and out-of-distribution (o.o.d.) prediction. The i.i.d. setting evaluates whether CellCast can accurately predict perturbation responses when training and test cells are sampled from the same data distribution. The o.o.d. setting evaluates whether the model can generalize to unseen datasets and species.

2.9.1 i.i.d. Setting

For the i.i.d. setting, we used single-cell perturbation datasets in which control and perturbed cells are available under matched cell types and perturbation conditions. Each dataset was randomly split into training, validation, and test sets with a ratio of 8:1:1. The training set was used for model optimization, the validation set was used for model selection and early stopping, and the test set was used only for final evaluation.

To reduce the influence of condition imbalance, control and perturbed cells were balanced within each cell type before training and evaluation. Specifically, for each cell type, we set

$$M = \min(N_{ctrl}, N_{pert}), \quad (16)$$

where N_{ctrl} and N_{pert} denote the numbers of control and perturbed cells, respectively. We then randomly sampled M cells from the condition with more cells. During prediction, only control cells from the test set were used as model input, while the corresponding perturbed cells were used only as ground truth for evaluation.

2.9.2 o.o.d. Setting

For the o.o.d. setting, we evaluated whether CellCast can generalize to perturbation responses that are not directly observed during training. In this setting, the model is trained on a subset of species and then tested on a held-out condition. The held-out condition is excluded from the training data and is used only for evaluation.

We first mapped genes from different species to a shared set of orthologous genes. CellCast was trained using control and perturbed cells from the source species and then applied to control cells from the target species. The predicted target-species perturbed expression profiles were compared with the observed target-species perturbed cells to examine whether CellCast can capture conserved perturbation response patterns across species.

3 Results

3.1 CellCast accurately predicts perturbation responses under the i.i.d. setting

We first evaluated CellCast on the COVID-19 perturbation dataset under the i.i.d. setting using 10 repeated 8:1:1 splits across six cell types. Following CoupleVAE, we used gene-wise mean expression R^2 , defined as the squared Pearson correlation between predicted and observed perturbed mean expression profiles, as the primary metric.

As shown in Figure 3A, CellCast achieved the best mean-expression prediction performance across all evaluated cell types. Averaged over six cell types and 10 repeats, CellCast obtained a mean R^2 of 0.9716, improving over CoupleVAE, MLP, and PCA-Ridge by 4.7%, 2.8%, and 4.4%, respectively. The gain was most evident in CD8 T, Secretory, and mDC, where baseline methods showed relatively weaker performance. In terms of unexplained variance ($1 - R^2$), CellCast reduced the error by 77.5% compared with CoupleVAE, indicating a substantially better recovery of the population-level perturbed transcriptional state.

We next examined whether the improved mean-expression prediction also translated into better perturbation response recovery. In the representative response-focused analyses shown in Figure 3B–D, CellCast consistently outperformed the baseline methods. On real top-100 DEGs, CellCast achieved the lowest mean absolute response error (0.1274), reducing the error by 55.4%

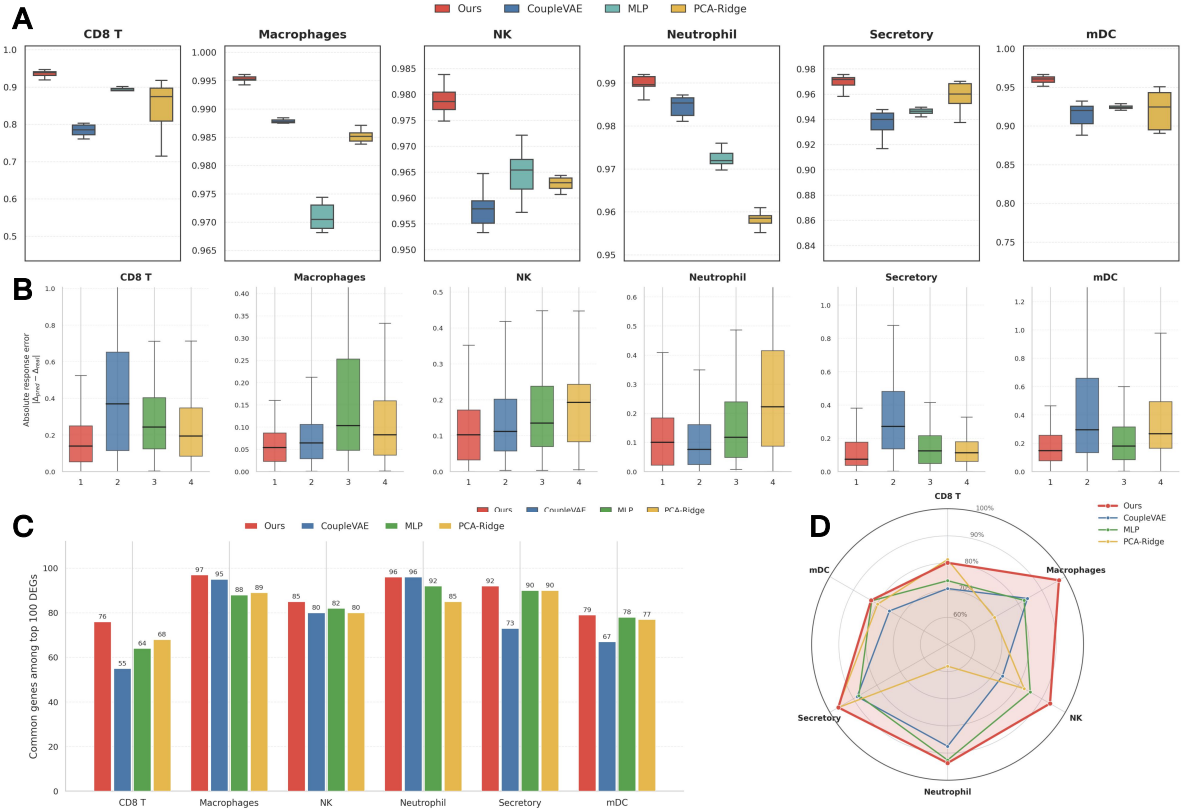


Figure 3 Performance of CellCast on the COVID-19 perturbation dataset under the i.i.d. setting. (A) Gene-wise mean expression R^2 across 10 repeated splits for six cell types. (B) Absolute perturbation response error on real top-100 differentially expressed genes. (C) Overlap between predicted and observed top-100 response genes. (D) Direction accuracy of predicted responses on top-1000 response genes.

compared with CoupleVAE and 34.7% compared with MLP. CellCast also recovered more true top-100 DEGs on average, with an overlap of 87.5 genes, corresponding to a 12.6% improvement over CoupleVAE. For response direction prediction, CellCast reached the highest average direction accuracy on the top-1000 response genes (0.9058), outperforming the strongest baseline by 7.9%. These results suggest that CellCast improves not only final expression prediction but also the recovery of perturbation-induced response magnitude, response genes, and response direction.

Together, these results demonstrate that CellCast provides a stronger population-level perturbation response predictor under the i.i.d. setting.

3.2 CellCast generalizes to cross-species perturbation prediction

We next evaluated whether CellCast can generalize perturbation response prediction across species. As illustrated in Figure 4A, the model was trained on source-species perturbation data and then applied to predict the LPS6-induced responses of a target species. Four target species, including mouse, rat, rabbit, and pig, were evaluated under the cross-species setting.

As shown in Figure 4B, CellCast achieved the highest mean-expression prediction accuracy for all four target species. Across five repeated runs, CellCast obtained an average R^2 of 0.9982, outperforming the strongest displayed baseline (0.9935). Although all methods achieved high

mean-expression agreement in this task, CellCast further reduced the unexplained variance ($1 - R^2$) by 72.3% relative to the strongest baseline, indicating more accurate recovery of target-species perturbed transcriptional states.

We further examined whether CellCast could recover perturbation-responsive genes in the target species. In the representative DEG recovery analysis shown in Figure 4C and Figure 4D, CellCast recovered an average of 93.75 real top-100 response genes, compared with 91.25, 90.75, and 88.00 for the baseline methods. CellCast also achieved the highest mean Jaccard similarity (0.8831), representing a 5.2% improvement over the strongest baseline. These results suggest that CellCast better preserves gene-level perturbation signals across species.

Finally, we visualized control, real perturbed, and predicted perturbed cells in a low-dimensional space for each target species. As shown in Figure 4E, the predicted perturbed cells were shifted from the control state toward the real perturbed state, providing qualitative support for cross-species

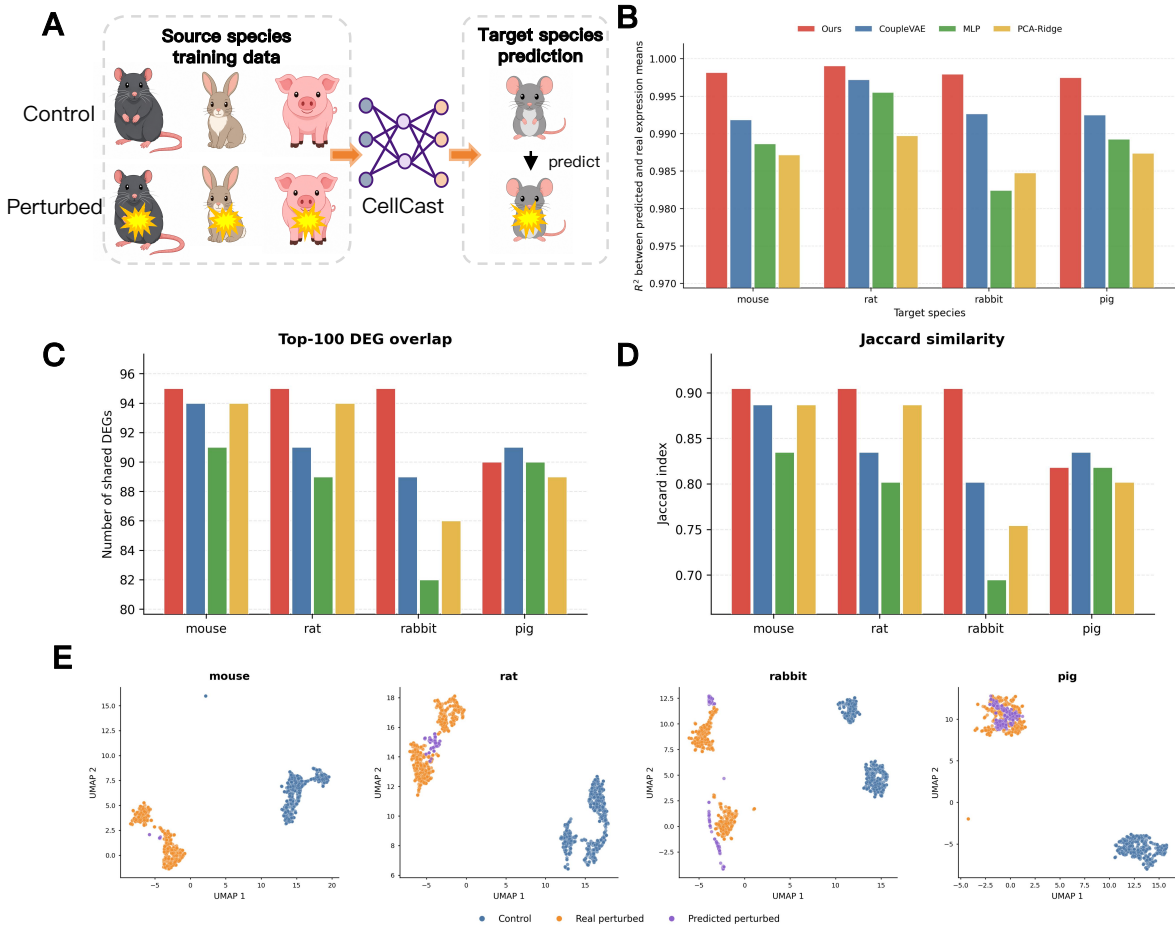


Figure 4 Cross-species perturbation prediction on the LPS6 dataset. (A) Schematic illustration of the cross-species setting, where CellCast is trained on source-species control and perturbed cells and then predicts perturbation responses in a target species. (B) Mean-expression R^2 between predicted and observed target-species perturbed profiles across mouse, rat, rabbit, and pig. (C) Recovery of target-species top-100 response genes, evaluated by DEG overlap and Jaccard similarity. (D) Jaccard similarity between predicted and observed top-100 response genes. (E) UMAP visualization of control, real perturbed, and predicted perturbed cells for each target species.

perturbation response transfer. Since this visualization is intended as a qualitative analysis, our main conclusion is based on the quantitative R^2 , DEG overlap, and Jaccard results.

Together, these results demonstrate that CellCast can transfer perturbation response patterns across species using shared orthologous genes. The improvement is reflected not only in mean-expression prediction accuracy, but also in better recovery of target-species perturbation-responsive genes.

4 Discussion

In this study, we proposed CellCast for AI-assisted single-cell perturbation response prediction. The main idea of CellCast is to predict perturbation-induced residual gene-expression changes from control cells, rather than generating perturbed expression profiles from scratch. This design is biologically reasonable because a perturbation usually changes part of the original cellular state instead of completely replacing it.

The i.i.d. results on the COVID-19 perturbation dataset showed that CellCast achieved strong prediction performance across multiple cell types. CellCast obtained the best average mean-expression R^2 of 0.9716, outperforming CoupleVAE, MLP, and PCA-Ridge. More importantly, the improvement was also observed in response-level analyses. CellCast achieved the lowest mean absolute response error on top DEGs, recovered more real top-100 response genes, and reached the highest response direction accuracy. These results suggest that CellCast does not only predict the final perturbed expression profile, but also better captures biologically meaningful perturbation responses.

The cross-species experiment further supports the generalization ability of CellCast. Using shared orthologous genes, CellCast achieved the highest mean-expression R^2 across mouse, rat, rabbit, and pig, with an average R^2 of 0.9982. It also recovered more target-species top-100 response genes on average. This indicates that CellCast can learn perturbation response patterns that are partially transferable across species. However, this result should be interpreted with biological caution, because cross-species prediction depends on shared genes and may not fully capture species-specific regulatory mechanisms.

These results also show why multiple evaluation metrics are necessary for perturbation prediction. A high expression-level R^2 indicates that the predicted perturbed profiles are close to the observed profiles, but it does not fully prove that the model has captured the perturbation mechanism. Therefore, response-level metrics such as DEG recovery, response magnitude error, and response direction accuracy are important. They help evaluate whether the model can identify which genes are changed by perturbation and whether these genes are predicted to change in the correct direction.

Despite these promising results, several limitations remain. First, the current experiments mainly support population-level perturbation response prediction, while single-cell-level heterogeneity is not fully validated. Second, the cross-species analysis relies on shared orthologous genes, which may ignore species-specific genes and regulatory programs. Third, CellCast is mainly data-driven and does not explicitly incorporate biological priors such as gene regulatory networks or pathways. In future work, more perturbation datasets, more species, biological prior knowledge, and wet-lab validation should be introduced to further improve the reliability and interpretability of CellCast.

5 Conclusion

In this work, we presented CellCast, an axial-attention-based model for single-cell perturbation response prediction. CellCast reformulates perturbation prediction as a residual gene-expression response prediction task, where the model learns perturbation-induced changes from control-state single-cell profiles. By organizing full-gene expression vectors into a computational gene grid and applying row-wise and column-wise attention, CellCast can model gene-token dependencies efficiently while preserving full-gene information.

Experiments on the COVID-19 perturbation dataset showed that CellCast achieved strong performance in the i.i.d. setting, with improved mean-expression prediction, response error reduction, DEG recovery, and response direction accuracy. In the cross-species LPS6 experiment, CellCast further demonstrated the ability to transfer perturbation response patterns across species using shared orthologous genes. These results suggest that CellCast can effectively capture population-level perturbation responses and generate useful computational hypotheses for biological research.

Overall, CellCast provides an AI-assisted framework for predicting single-cell perturbation responses before performing costly laboratory experiments. Although further validation on more datasets and wet-lab experiments is still needed, this study shows the potential of combining deep learning with single-cell perturbation biology to support hypothesis generation and experimental prioritization.

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